



Introduction to Enterprise Information Systems (EIS) and the Strategic Importance of IT

Business Information Systems • Weeks 01–02

Cyber.SoHo Educational Hub

From the Industry to the Classroom — Building the IT's Future, Today and Together!

About this deck



- This is the **projector companion** to the full lecture packet.
- It carries the **concepts, frameworks, and formulas** only.
- The worked code and the long stories live in the **lecture**, so the room can **watch the screen**, not read it.
- Designed for **first-time instructors** and for **asynchronous self-study**.

KEY TAKEAWAY

Slides for the room; the lecture packet for the detail. Use them together.

Agenda



Lesson 1 — Introduction to Enterprise Information Systems (EIS)

- 1 The six building blocks of an information system
- 2 The four layers — the information systems pyramid
- 3 What “enterprise” really means: integration

Lesson 2 — The Strategic Importance of IT

- 1 Porter's Five Forces, and how IT answers each one
- 2 Porter's Value Chain, and where IT adds value
- 3 Putting a number on it: the business value of IT

KEY TAKEAWAY

Lesson 1 asks what an enterprise system IS. Lesson 2 asks why it MATTERS. Same machine, two questions.

Learning objectives



By the end of these two lessons you should be able to:

- **Name** the six parts every information system is built from.
- **Place** a real system on the four-layer pyramid (TPS → ESS).
- **Explain** the difference between scattered data silos and one integrated EIS.
- **Apply** Porter's Five Forces and Value Chain to a familiar company.
- **Calculate** whether an IT project is worth the money, using payback and ROI (Return On Investment).

KEY TAKEAWAY

You are not here to memorise vocabulary — you are here to USE five frameworks on companies you already know.



Lesson 1: Introduction to Enterprise Information Systems (EIS)

What an enterprise system is — end to end

What is an Enterprise Information System?



- An **IS** (Information System) collects, stores, processes, and shares information to help people make decisions.
- An **EIS** (Enterprise Information System) is an IS that serves a **whole organisation at once** — every department working from the **same** data.
- The defining word for “enterprise” is **integration**.

KEY TAKEAWAY

“Enterprise” is not about size. It is about one shared source of truth instead of many private copies.



Point 1.1 — The six building blocks of an IS (1/2)

Every information system — from a coffee-shop till to an airline — is built from the same six parts:

- 1 **Hardware** — the physical devices (screens, servers, scanners).
- 2 **Software** — the instructions that tell the hardware what to do.
- 3 **Data** — the raw facts the system records and stores.
- 4 **Networks** (Telecommunications) — the wiring and signals that move data.
- 5 **People** — the users and specialists who run and rely on the system.
- 6 **Processes** (Procedures) — the agreed steps that turn the rest into useful work.



Point 1.1 — The six building blocks of an IS (2/2)

Why it matters

- Remove **any one** block and the system stops being useful.
- A great app (**software**) with no trained staff (**people**) or no agreed steps (**processes**) still fails.

KEY TAKEAWAY

Technology is only two of the six blocks. PEOPLE and PROCESSES decide whether the technology ever pays off.

Point 1.2 — The information systems pyramid (1/2)



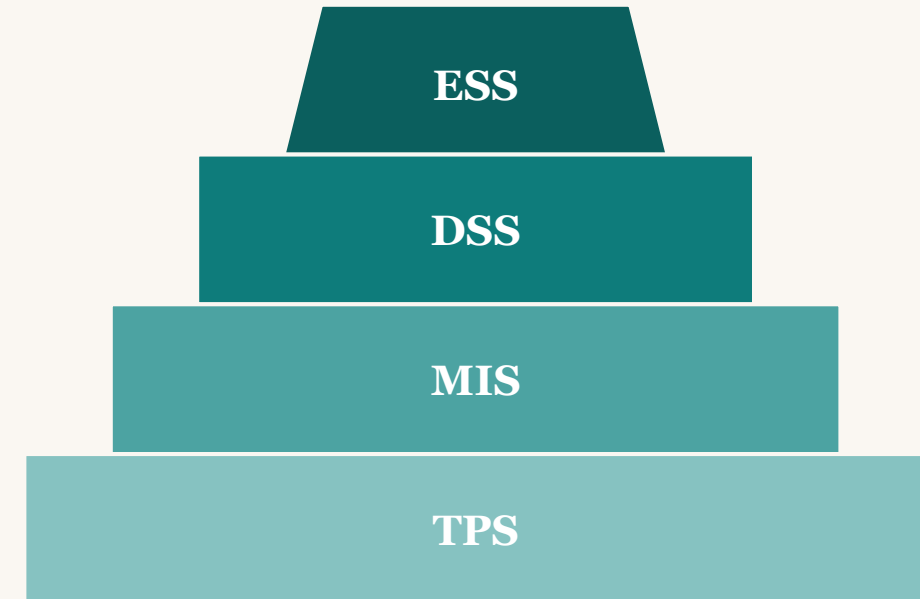
Organisations stack systems by WHO uses them and WHAT kind of decision they support:

Layer	Serves	Typical question it answers
TPS — Transaction Processing System	Front-line / operations	“Did this sale go through?”
MIS — Management Information System	Middle managers	“How did the store do this week?”
DSS — Decision Support System	Analysts / managers	“What if we change the price?”
ESS — Executive Support System	Senior executives	“Where should the company go next?”



Point 1.2 — The information systems pyramid (2/2)

- The **TPS** at the base captures **every individual event** (one scan, one payment).
- Each layer above **summarises** the one below for a **wider view** and a **longer horizon**.
- Detail shrinks as you climb; **strategy grows**.



raw transactions in at the base → strategy off the top

KEY TAKEAWAY

Raw transactions enter at the bottom; strategy is read off the top — one continuous flow of the same data, zoomed out at each step.



Point 1.3 — What “enterprise” means: integration (1/3)

Data silos — the problem

- Each department keeps its **own private copy** of the data.
- Copies **drift apart** — different addresses, different totals, different “truth.”
- Result: **conflict, rework**, and decisions based on **stale numbers**.

KEY TAKEAWAY

Silos store the same fact many times — and let the copies quietly disagree.



Point 1.3 — What “enterprise” means: integration (2/3)

One integrated EIS — the fix

- Every department points at **one shared record**.
- Update it once → **everyone** sees the change instantly.
- No more reconciling copies; the data is **consistent by design**.

KEY TAKEAWAY

An EIS stores each fact ONCE and lets everyone share it. That single difference is the whole point of “enterprise.”



Point 1.3 — Enterprise applications that integrate (3/3)

Application	Pulls together
ERP — Enterprise Resource Planning	Finance, operations, inventory, HR (Human Resources)
CRM — Customer Relationship Management	Every customer interaction across sales and support
SCM — Supply Chain Management	Suppliers, logistics, and delivery
KMS — Knowledge Management System	The organisation's documents, know-how, best practices

KEY TAKEAWAY

ERP, CRM, SCM, and KMS are four doors into the SAME integrated data — each serving a different slice of the business.



Lesson 1 — watch this (optional, under 15 minutes)



- **Introduction to Enterprise Information Systems** — youtube.com/watch?v=JZTKRwHBkoM
- Gentle warm-up alternative — youtube.com/watch?v=g8059-8QDNk

Instructor note: confirm the link plays **before class** and pass any URL through **VirusTotal** (virustotal.com/gui/home/url). Link to the video — never re-host or download the creator's work. That protects their author rights and yours.

KEY TAKEAWAY

Always link out to the original creator; verify the URL first. Respect author rights every time.



In one breath

KEY TAKEAWAY

Six building blocks → stacked into four layers → integrated into one enterprise system. That is WHAT it is, end to end.



Lesson 2: The Strategic Importance of IT

Why IT matters — end to end

The question on the table (1/2)



- In 2003, Nicholas Carr published “**IT Doesn't Matter**” in **HBR** (Harvard Business Review).
- His argument: once **everyone** has the same technology, it stops being an advantage — **like electricity**.

KEY TAKEAWAY

Carr's challenge: if every rival can buy the same technology, how can technology be an advantage at all?

The question on the table (2/2)



The counter-argument (and this lesson)

- The **technology** may be a commodity — but **how you use it** is not.
- The strategy is in the **use**, not the box.

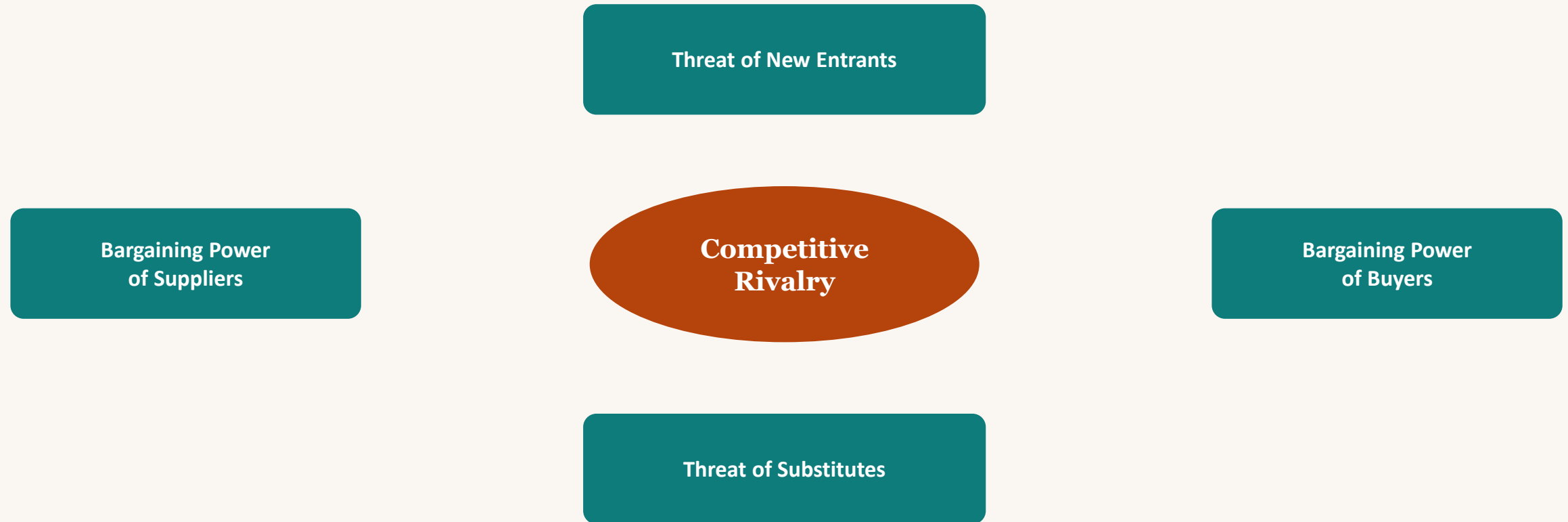
KEY TAKEAWAY

IT becomes STRATEGIC the moment it changes what the business can do that rivals cannot.

Point 2.1 — Porter's Five Forces (1/3)



Michael Porter's framework explains why some industries are profitable and others brutal — and IT gives a lever on every force.



Point 2.1 — Five Forces & the IT lever (2/3)



Force	What it asks	An IT lever
Competitive Rivalry	How fierce is the fight already?	Data + automation to undercut on cost or speed
Threat of New Entrants	How easily can newcomers attack?	Scale and a strong platform raise the barrier
Threat of Substitutes	Could a different product replace yours?	Use data to keep improving the offer
Power of Suppliers	Can suppliers squeeze you?	Connect many suppliers; cut dependence on one
Power of Buyers	Can customers force prices down?	Personalisation and lock-in raise switching costs

Point 2.1 — How to read the Five Forces (3/3)



- **Strong** forces → thin profits, hard industry.
- **Weak** forces → room to earn and grow.
- IT does not remove a force; it **shifts the balance** in your favour.

KEY TAKEAWAY

Five Forces is a DIAGNOSIS of an industry. IT is one of the strongest TREATMENTS you can apply to each force.



Point 2.2 — Porter's Value Chain (1/3)

A company is a chain of activities, each adding value on the way to the customer. PRIMARY activities do the core work:

- 1 **Inbound logistics** — receiving and storing inputs
- 2 **Operations** — turning inputs into the product
- 3 **Outbound logistics** — delivering to the customer
- 4 **Marketing & Sales** — finding and winning buyers
- 5 **Service** — supporting the customer afterwards



Point 2.2 — Support activities & the value formula (2/3)

Support activities (hold it all up)

- Firm infrastructure, HR (Human Resources) management, Technology development, Procurement.

$$\begin{aligned}\text{Margin} &= \text{Value} - \text{Cost} \\ \text{Margin \%} &= (\text{Margin} \div \text{Value}) \times 100\end{aligned}$$

KEY TAKEAWAY

Every activity adds value and incurs cost. The gap left over is the MARGIN — the whole point of the chain.

Point 2.2 — Where IT widens the margin (3/3)



- **Value** = what the customer will pay.
- **Cost** = what the activity costs you.
- IT works to **raise value** (better product, faster service) **or lower cost** (automation) — widening the **margin**.

KEY TAKEAWAY

Good IT pushes value UP and cost DOWN at the same time — that is how technology becomes profit.



Point 2.3 — The business value of IT: payback (1/3)

A project is only “strategic” if the numbers agree. Two tools decide it — the first is payback period.

**Payback = time until cumulative savings
repay the original investment**

- Add up the **net yearly gain** until the running total **turns positive**.
- **Shorter payback = lower risk.**

KEY TAKEAWAY

Payback answers HOW SOON you break even — the faster the money comes back, the safer the project.

Point 2.3 — Return On Investment & the model (2/3)



$$\text{ROI \%} = (\text{Net gain} \div \text{Amount invested}) \times 100$$

Metric	Value
Initial investment	\$40,000 CAD
Annual saving	\$18,000
Annual running cost	\$3,000
Net yearly gain	\$15,000
Payback period	≈ 2.7 years
5-year net gain	\$35,000
5-year ROI	87.5 %

Point 2.3 — The verdict in three numbers (3/3)



\$15k

net gain per year

2.7 yr

payback period

87.5%

five-year ROI

KEY TAKEAWAY

“It feels worth it” is not a business case. Payback shows HOW SOON; ROI shows HOW MUCH. Decisions need both numbers.



Lesson 2 — watch this (optional, under 15 minutes)



- **Porter's Five Forces Explained** (~12 min) — youtube.com/watch?v=Dfp23xSqpdK
- Short alternative — youtube.com/watch?v=IPHruQHAECw

Instructor note: verify playback before class and scan the URL through **VirusTotal**. Link out to the creator's video — do not re-host or download it.

KEY TAKEAWAY

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In one breath

KEY TAKEAWAY

Diagnose the industry (Five Forces) → find where you add value (Value Chain) → prove it pays (payback + ROI). That is WHY IT matters, end to end.

Bringing it together



- **Lesson 1** built the machine: six blocks → four layers → one integrated EIS.
- **Lesson 2** aimed the machine: weaken the forces, widen the margin, prove the return.
- A system is only **strategic** when the **business value** (Lesson 2) flows out of the **integration** (Lesson 1).

KEY TAKEAWAY

Architecture without strategy is expensive plumbing. Strategy without architecture is a wish. The two lessons are ONE idea.

Key abbreviations (1/2)



Short	In full
EIS	Enterprise Information System
IS	Information System
IT	Information Technology
TPS	Transaction Processing System
MIS	Management Information System
DSS	Decision Support System
ESS	Executive Support System

Key abbreviations (2/2)



Short	In full
ERP	Enterprise Resource Planning
CRM	Customer Relationship Management
SCM	Supply Chain Management
KMS	Knowledge Management System
ROI	Return On Investment
KPI	Key Performance Indicator
HR	Human Resources

KEY TAKEAWAY

Every term is defined in full in the lecture glossary (27 entries) and in `data/ref_glossary-abbreviations.csv`.

Links & resources



- The **full lecture packet** carries every clickable link, each with a one-line explanation.
- **Frameworks credited to their authors:** Porter's Five Forces and Value Chain → Michael Porter / **HBR** (hbr.org) and the Institute for Strategy & Competitiveness (isc.hbs.edu).
- **Verify any URL before opening it:** VirusTotal — [virustotal.com/gui/home/url](https://www.virustotal.com/gui/home/url).

KEY TAKEAWAY

Always send students to the ORIGINAL source. Attribute frameworks to their authors and link out — never copy.

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